

Problem 3.2

Solution:

(a)

(i) Subproblems:

Define $C(i, k)$ to be the minimum cost of a path to u using at most k edges.Want $\min_{v \in V} \{C(v, K) + p(v)\}$.

(ii) Recursive formula:

$$C(i, k) = \min_{e \in E} \{C(i, k), C(j, k-1) + w(e)\}$$

where e is an edge from j to i .

Base case:

$$C(i, 0) = \begin{cases} 0, & \text{if } i = s, \\ +\infty, & \text{otherwise.} \end{cases}$$

(iii) Evaluation order:

Increase k from 1 to K , for each k , iterate through all $e \in E$ to evaluate C .

(iv) Optimal cost:

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C = array of size (n, 2) initialized to inf
C[s][0] = 0
P = array of size (n, K) recording previous step
for k = 1 to K:
    for e in E:
        # suppose e is an edge from j to i
        evaluate C[i][1] according to (ii)
        update P if there is a better path to i
    roll C
for v in V:
    for k = 1 to K:
        find min cost according to (i)

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(v) Optimal path:

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v = vertex with minimum cost from (iv)
path = array storing the optimal path

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for k = K to 1:
    add P[v][k] to the beginning of path
    v = P[v][k]
add s to the beginning of path
return path

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(vi) Time and space complexity:

Iteration takes $O(mK)$ steps, each step takes $O(1)$ time to evaluate. Finding minimum cost takes extra $O(nK)$ time, which is smaller than $O(mK)$.

Total time complexity is (mK) .

Total space complexity is (nK) .

(b) Define $D(L, R, C_L) = C_R$ to be the recursive routine with state C_L from step L to R where $1 \leq L < R \leq K$, $P_k(i)$ to be the parent pointer at step k .

One observation is we do not need $P_{i+\Delta}$ or $C_{i+\Delta}$ ($\Delta > 0$) while reconstructing step i .

Suppose $mid = \frac{L+R}{2}$, recursively compute $D(L, mid, C_L)$ then $D(mid, R, C_{mid})$, record parent pointer $P_{mid}(i)$ and C_{mid} on the right side only (e.g. $C_8, C_{12}, C_{14}, C_{15}$ if $n = 16$), taking $O(n \log K)$ space in total.

After computing the optimal cost, reconstruct path according to P and C recorded at each recursive depth, recompute left sides recursively to replace P and C on the right side (e.g. replace C_{15} with C_{13} at depth 4), in order to reconstruct full path on both sides. Time complexity is $O(nK)$ at each depth. Total time complexity is $O(nK \log K)$.